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ANUDIP FOUNDATION

IT MODULE

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DEFINITION OF INFORMATION TECHNOLOGY

- **What is IT?** The study, design, development, implementation, and management of computer-based information systems.
- **The Scope:** It encompasses both physical equipment (hardware) and the invisible instructions (software) used to manage data.
- **Key Objective:** To convert raw **Data** into meaningful **Information** that supports human decision-making.

FUNDAMENTAL & CORE CONCEPTS OF IT

- **The Information Chain:**
- 1. **Input:** Capturing raw data.
- 2. **Processing:** Converting data into a useful format.
- 3. **Output:** Distributing information to users.
- 4. **Storage:** Retaining data for future use.
- **Connectivity:** The concept of Networking (LAN/WAN) and the Internet as the backbone of global communication.

INTRODUCTION TO IT (THE "BIG PICTURE")

Evolution: From mainframes and physical servers to Cloud Computing and Edge Computing.

•Ubiquity: IT is no longer just a department; it is the "nervous system" of every industry (Healthcare, Finance, Agriculture, Education).

•Drivers of IT: Automation, Scalability, and Global Connectivity.

DIFFERENCE BETWEEN HARDWARE AND SOFTWARE

Feature	Hardware	Software
Nature	Physical components (Tangible)	Set of instructions/programs (Intangible)
Function	Executes tasks and processes data	Directs the hardware on what to do
Failure	Wears out over time (Depreciation)	Does not wear out; becomes obsolete/bugs
Examples	CPU, RAM, Hard Drive, Sensors	OS (Windows/Linux), ERP, Java Apps

CORE IT FUNCTIONS (PART A): DATA BACKUP & RECOVERY

- **Data Backup:** The process of creating a copy of data to protect against loss (Full, Incremental, or Differential backups).
- **Disaster Recovery (DR):** The strategy for restoring IT infrastructure and operations after a crisis (e.g., cyberattack or hardware failure).
- **Importance:** Ensures **Business Continuity**; without a recovery plan, data loss can lead to permanent business closure

CORE IT FUNCTIONS (PART B): NETWORK & VIRTUALIZATION

- **Network Management:** Monitoring and maintaining the "traffic" of data between computers to ensure speed and security.

- **Virtualization Software:** * **Definition:** Technology that allows you to create multiple simulated environments (Virtual Machines) from a single physical hardware system.
- **Benefits:** Reduces hardware costs, improves server utilization, and allows for easier testing of software.

THE ROLE OF IT IN MODERN BUSINESS

1. Operational Efficiency: Automating repetitive tasks to save time and reduce human error.
2. **Remote Collaboration:** Enabling global teams to work together via cloud tools and video conferencing.
3. **Security:** Protecting intellectual property and customer data through cybersecurity protocols.

INTRODUCTION TO ENTERPRISE RESOURCE PLANNING (ERP)

- Concept: A software system that helps businesses integrate and manage their core business processes in a single, unified system.
- The Problem it Solves: Before ERP, departments (Finance, HR, Sales) used separate "silos" of data that didn't talk to each other.
- The Result: A "Single Source of Truth" for the entire organization.

DEFINITION & WORKING OF ERP

•**Definition:** A suite of integrated applications used by an organization to collect, store, manage, and interpret data from many business activities.
•**How it Works:** *
• **Centralized Database:** All modules feed into one database.
•**Real-time Processing:** When a sale is made, inventory is updated instantly.
•**Automation:** Triggers automatic actions (e.g., generating an invoice when a product is shipped).

THE 5 CORE COMPONENTS OF ERP

- Hardware: Servers, cloud infrastructure, and client devices.
- Software: The actual ERP application and operating systems.
- Data: The information stored within the system (Customer, Financial, Inventory).
- People: The users (employees) and IT staff who manage the system.
- **Procedures:** The business rules and workflows that define how the software is used.

IMPORTANCE & BENEFITS OF ERP

Efficiency: Reduces manual data entry and repetitive tasks.

- Visibility: Provides managers with real-time dashboards to see business health.
- Compliance: Built-in tools for financial reporting and regulatory standards (e.g., GST or GDPR).
- Cost Savings: Lower operational costs through better resource planning.

ADVANTAGES & DISADVANTAGES OF ERP

- **Advantages:**
- Improved data security and consistency.
- Enhanced collaboration across departments.
- Better customer service through faster order fulfillment.
- **Disadvantages:**
- High initial cost (Software licenses + Implementation).
- Complexity: Requires extensive employee training.
- Resistance to Change: Employees may struggle with new digital workflows.

TYPES OF ERP (CATEGORIZATION BY DEPLOYMENT)

- **On-Premise ERP:** Installed locally on the company's own hardware/servers.
- **Cloud ERP (SaaS):** Hosted on the provider’s servers; accessed via a web browser.
- **Hybrid ERP:** A mix of on-premise and cloud solutions to balance security and flexibility.
- **Open Source ERP:** Allows organizations to access and modify the source code for custom needs.

DETAILED CATEGORIZATION OF ERP SYSTEMS

- **Generalist ERP:** Designed for many industries with broad features (e.g., SAP, Oracle).
- **Vertical (Industry-Specific) ERP:** Built for specific niches like Manufacturing, Construction, or Healthcare.
- **Small Business ERP:** Scaled-down versions with essential features (e.g., Tally Prime, Zoho Books).

ERP VS. CRM: STRATEGIC DIFFERENCE

- **ERP (Back-Office):** Focuses on internal processes like Finance, HR, and Supply Chain.
- **CRM (Front-Office):** Focuses on external relationships, sales leads, and customer support.
- **Integration:** Ideally, a CRM feeds customer data into the ERP to manage the actual fulfillment of orders.

EXAMPLES OF MODERN ERP SYSTEMS

- **Tier 1 (Large Enterprise):** SAP S/4HANA, Oracle ERP Cloud, Microsoft Dynamics 365.
- **Tier 2 (Mid-Market):** Infor, Sage Intacct, NetSuite.
- **Tier 3 (Small Business):** Odoo, Tally, Zoho.

RELATIONSHIP BETWEEN BUSINESS STRATEGY & DATA

- **Strategy Alignment: IT systems must be designed to support the specific goals of the company (e.g., being the lowest-cost provider vs. the highest-quality provider).**

- **Data-Driven Strategy: Modern businesses use ERP data to pivot their strategy based on market demand.**

- **KPIs: Strategy is measured through Key Performance Indicators derived directly from the ERP database.**

COMMON DATA SOURCES FOR ORGANIZATION

- Internal Sources: ERP logs, CRM records, employee feedback, financial statements.
- External Sources: Market research, social media trends, government regulations, competitor pricing.
- IoT (Internet of Things): Sensors on factory floors or delivery trucks providing real-time data.

DATA ASSETS

- **The New Oil:** In modern business, data is an asset just like physical inventory or cash.
- **Objective:** To understand how raw facts are captured, organized, and turned into strategic insights.
- **Key Focus:** Collection methods, metadata, and the transformation process.

DEFINING DATA & METADATA

- **What is Data?**
- Raw, unorganized facts and figures (numbers, text, images) that have no meaning on their own.

- **What is Metadata?**
- Often described as **"Data about Data.“**
- **Example:**
- If a digital photo is the "Data," then the date taken, file size, and GPS coordinates are the "Metadata."
- **Role of Metadata:**
- It provides context, making data searchable, manageable, and understandable within an IT system.

CHARACTERISTICS OF QUALITY DATA

- **Accuracy:** Is the data correct?
- **Completeness:** Is any critical information missing?
- **Consistency:** Does the data match across different systems (e.g., ERP vs. CRM)?
- **Timeliness:** Is the data up-to-date for current decision-making?

DATA COLLECTION: DEFINITION & PROCESS

- **Definition:** The systematic approach to gathering and measuring information from various sources.
- **The 4-Step Process:**

1. **Identify:** Determine what data is needed for the business goal.
2. **Gather:** Use specific tools to collect the raw facts.
3. **Store:** Save the data in a secure database or cloud environment.
4. **Clean:** Remove duplicates or errors before analysis

METHODS & TOOLS FOR DATA COLLECTION

•**Primary Methods:** * **Surveys/Questionnaires:** Direct feedback from customers.
•**Interviews:** In-depth qualitative data.
•**Observations:** Monitoring user behavior on a website.
•**Modern Tools:**
•**Web Scrapers:** (e.g., Beautiful Soup, Octo parse) for gathering online market data.
•**IoT Sensors:** For real-time machine performance data.
•**CRM Logs:** Tracking every interaction a customer has with the brand.

DATA VS. PROCESSED DATA

- **The Transformation**
- **Raw Data:** "100, 150, 200" (Meaningless numbers).
- **Processed Data:** "Daily Sales Revenue for Monday, Tuesday, Wednesday."
- **The Transformation:** * **Sorting:** Organizing data numerically or alphabetically.
  - **Summarizing:** Calculating averages or totals.
  - **Aggregating:** Combining data from multiple sources to see the "Big Picture.

TRANSFORMING INFORMATION INTO KNOWLEDGE

•**Information:** Organized data that answers "Who, What, Where, and When."
•**Knowledge:** Applying experience and analysis to information to answer **"How."**
•**Example in IT:** * **Data:** A list of customer ages.
•**Information:** A chart showing 70% of customers are under age 25.
•**Knowledge:** Realizing our marketing should focus on social media (Instagram/TikTok) rather than newspapers.

INTRODUCTION TO DATA LIFECYCLE MANAGEMENT (DLM)

- **The "Cradle-to-Grave" Approach:** DLM is a policy-based approach to managing the flow of an information system's data.
- **The Core Need:** Not all data is equally valuable forever. DLM helps decide what to keep on fast servers and what to archive or delete.
- **Scope:** It involves the business processes, policies, and software used to manage data phases.

DEFINITION, GOALS, AND IMPORTANCE

•**Definition:** A set of procedures for managing data throughout its entire existence, ensuring it remains accessible, secure, and accurate.
•**Three Primary Goals:**
1. **Data Confidentiality:** Protecting sensitive information.
2. **Availability:** Ensuring data is ready when needed for business decisions.
3. **Integrity:** Keeping data accurate and free from corruption.

•**Importance:** It reduces storage costs and ensures the company meets legal requirements (like keeping financial records for 7 years).

THE 5 STANDARD STAGES OF DLM

- **Creation/Collection:** Data enters the system (e.g., a customer fills out a form).
- **Storage/Maintenance:** Data is saved in a secure database with regular backups.
- **Usage:** Data is actively used for reports, analytics, or business operations.
- **Archival:** Data is moved to long-term, cheaper storage (it's rarely used but must be kept).
- **Destruction/Deletion:** Data is permanently and securely erased once its legal/business life ends.

STEP 1: DATA CLASSIFICATION: LABELING DATA AS "PUBLIC," "INTERNAL," OR "CONFIDENTIAL.

- **Step 2: Policy Development:** Creating rules for how long each type of data should be stored.
- **Step 3: Automation:** Using software to move data from "Hot Storage" to "Cold Storage" automatically based on age.
- **Step 4: Monitoring:** Auditing the lifecycle to ensure no data is leaking or being kept illegally.

TOOLS FOR EFFECTIVE DLM

- **Storage Management Tools:** Software that manages tiered storage (e.g., AWS S3 Lifecycle policies).
- **Data Masking Tools:** To protect sensitive data during the "Usage" stage.
- **Backup & Recovery Suites:** (e.g., Veeam, Veritas) to manage the "Storage" stage.
- **Secure Erase Tools:** For the "Destruction" stage to ensure data cannot be recovered by hackers.

CHALLENGES OF DLM

- **Data Volume:** Managing the sheer "Big Data" produced every second.
- **Regulatory Complexity:** Different countries have different laws for data retention (e.g., GDPR in Europe).
- **Siloed Data:** Data stuck in different departments (HR vs. Finance) makes a unified policy difficult.
- **Security Risks:** The "Archival" stage is often overlooked and becomes a target for hackers.

DLM VS. ILM (INFORMATION LIFECYCLE MANAGEMENT)

- **DLM (Data-Centric):** Manages the **file** or the "bucket of data" based on its age, size, and type.
- **ILM (Information-Centric):** A more complex approach that manages the **content** of the data.
- *Example:* If a file contains a specific keyword like "Contract," ILM treats it differently than a standard email, even if they are the same age.
- **Key Difference:** DLM is about **Storage Management**; ILM is about **Record Management.**

DATA INTEGRITY & QUALITY ASSURANCE

- **The "Garbage In, Garbage Out" (GIGO) Principle:** If raw data is incorrect or corrupted, even the most expensive ERP system will produce wrong results.
- **Goal:** To ensure data is not only correct for its intended use (**Quality**) but also remains unchanged and reliable over time (**Integrity**).

DATA INTEGRITY VS. DATA QUALITY: DETAILED ANALYSIS

•**Data Integrity (The "Wholeness" of Data):**
•Refers to the **validity and consistency** of data over its entire lifecycle.
•It focuses on protecting data from unauthorized changes or accidental corruption.
•**Data Quality (The "Utility" of Data):**
•Refers to how well the data serves its **specific purpose**.
•It focuses on accuracy, completeness, and reliability for business operations.
•**Key Difference:** Integrity is a **technical state** (Is the file broken?); Quality is a **business state** (Is the customer's phone number correct?).

DETAILS OF DATA INTEGRITY (TYPES & MECHANISMS)

•**Physical Integrity:** Protecting data against hardware failures, power outages, or natural disasters.
•**Logical Integrity:** * **Entity Integrity:** Ensuring every record has a unique ID (Primary Key).
•**Referential Integrity:** Ensuring relationships between tables (e.g., an Order must belong to an existing Customer).
•**Domain Integrity:** Ensuring data follows set rules (e.g., a "Date" field cannot contain "ABCD").

DETAILS OF DATA QUALITY (THE 5 DIMENSIONS)

- **Accuracy:** Does it represent the real-world value?
- **Completeness:** Are there any "Null" or empty fields in critical areas?
- **Consistency:** Is the data the same across the ERP, CRM, and Website?
- **Reliability:** Can the data be trusted to yield the same results over time?
- **Timeliness:** Is the data current enough to be useful? (e.g., yesterday's stock prices are useless for today's trade).

BENEFITS OF GOOD DATA INTEGRITY & QUALITY

- **Informed Decision-Making:** Executives can trust reports without second-guessing the numbers.
- **Regulatory Compliance:** Essential for audits (SOX, HIPAA, or Banking regulations).
- **Enhanced Customer Trust:** Preventing errors like double-billing or shipping to the wrong address.
- **Operational Efficiency:** Employees spend less time "fixing" data and more time using it.
- **Example:** A bank maintaining **Data Integrity** ensures that when you transfer $100, it is deducted from one account and added to the other simultaneously without a "glitch."

BUSINESS ANALYSIS & AGILE METHODOLOGIES

- **Introduction to Business Analysis**
- **Definition:** The practice of enabling change in an organizational context by defining needs and recommending solutions that deliver value to stakeholders.
- **The Bridge:** Business Analysis acts as the bridge between **Business Stakeholders** (who have a problem) and **IT Teams** (who build the solution).

SLIDE 40: PRINCIPLES, TOOLS & APPLICATIONS

- **Core Principles:** 1. Understand the Business Need. 2. Define the Future State. 3. Assess the Gap.
- **Applications:** Product Development, Process Improvement, and Strategic Planning.
- **Essential Tools:** * **Modeling:** Microsoft Visio, Lucid chart (for Flowcharts/BPMN).
- **Requirement Tracking:** Jira, Trello, Confluence.
- **Data Analysis:** Microsoft Excel, Tableau, Power BI

BUSINESS ANALYST VS. BUSINESS ANALYTICS

- **Business Analyst (The Person):** Focuses on processes, requirements, and communication. They ask, "What does the business need?"
- **Business Analytics (The Process):** Focuses on data, statistical models, and trends. It asks, "What does the data tell us?"
- **Integration:** A great Business Analyst uses Business Analytics to prove their recommendations with data.

FOUR TYPES OF ANALYTICS

- **Descriptive (What happened?):** Summarizing past data (e.g., Monthly Sales Reports).
- **Diagnostic (Why did it happen?):** Finding the root cause (e.g., Why did sales drop in June?).
- **Predictive (What will happen?):** Using trends to forecast the future (e.g., Sales projections for Q4).
- **Prescriptive (How can we make it happen?):** Recommending a specific action based on the forecast.

SOFTWARE DEVELOPMENT MODELS: WATERFALL VS. AGILE

•**Waterfall Model:** A linear, sequential approach. You must finish one phase (Requirements) before moving to the next (Design). Best for fixed-scope projects.

•**Agile Technology:** An iterative approach. Work is broken into small "Sprints" (2–4 weeks). It allows for constant feedback and changes. [Image comparing Waterfall model vs Agile iterative cycle]

WATERFALL VS AGILE

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	Waterfall	Agile
Requirements	Clearly-defined	Can change
Client involvement	×	✓
Type	Straightforward solutions	Complex projects
Time frames	Strict	Loose
Budget	Strict	Big or unlimited
Team collaboration	Moderate	Intensive

STAKEHOLDER MANAGEMENT & INFORMATION FOR DECISIONS

- **Stakeholder Management:** Identifying everyone affected by a project and keeping them satisfied through clear communication.
- **Benefits:** Reduces resistance to change and ensures project requirements are accurate.
- **Transforming Info into Decisions:** How "Processed Data" becomes the evidence used in Boardroom meetings to approve budgets or new products.

LEADERSHIP, MANAGEMENT & ETHICS

- **Management Fundamentals: Span of Management**
- **Definition:** Also called "Span of Control," it refers to the number of subordinates a manager can effectively supervise.
- **Factors Affecting Span:** * Complexity of the work.
  - Competence of the employees.
  - Degree of centralization/automation in the IT system.

EMPLOYEE MORALE & JOB SATISFACTION

- **Employee Morale:** The overall outlook and attitude of employees toward their workplace.
- **Factors Affecting Morale:** Leadership style, workload, and career growth opportunities.
- **Job Satisfaction Factors:** Fair pay, work-life balance, and feeling that their work (especially in IT) has a purpose.
- **Overcoming Low Morale:** Transparent communication, rewarding achievements, and providing the right IT tools to reduce frustration.

DATA-DRIVEN DECISION MAKING (DDDM)

- **Definition:** Making choices based on data analysis rather than intuition or "gut feeling."
- **Transitioning Benefits:** Increased transparency, higher success rates, and measurable ROI.
- **5 Steps for Implementing DDDM:**
- 1. Define goals.
- 2. Identify data sources.
- 3. Collect/Clean data.
- 4. Analyze.
- 5. Act on insights

DATA GOVERNANCE & ETHICAL PRINCIPLES

- **Data Governance:** The overall management of the availability, usability, integrity, and security of data in an enterprise.
- **Ethical Data Collection:** 1. **Transparency:** Tell users you are collecting data. 2. **Consent:** Get permission. 3. **Anonymization:** Protecting individual identity in large datasets.

REGULATION & CULTURE (CCPA VS. GDPR)

- **GDPR (Europe):** Very strict; gives users the "Right to be Forgotten."
- **CCPA (California):** Focuses on the right to know what data is sold and to opt-out.
- **Data Controller Culture:** Shifting a company from "owning" data to being a "steward" of data.
- **Pitfalls:** Avoiding "Scenario Analysis" errors and policy gaps that lead to data breaches.

CONCLUSION

- **Integrated Ecosystem:** Information Technology is no longer a support function; it is the **backbone of modern business strategy**, seamlessly integrating hardware, software, and human resources through systems like **ERP**.
- **Data as a Strategic Asset:** The transition from raw **Data** to **Information** and finally **Knowledge** is the most critical value-addition process in any organization.
- **Lifecycle & Quality:** Implementing robust **Data Lifecycle Management (DLM)** and ensuring **Data Integrity** are non-negotiable for operational efficiency and regulatory compliance.
- **Agility in Analysis:** Moving from traditional **Waterfall** models to **Agile technologies** allows businesses to remain competitive and responsive to stakeholder needs in a fast-paced market.
- **Ethical Leadership:** The future of IT lies in **Data-Driven Decision Making (DDDM)** balanced with strong **Ethical Principles** (like GDPR/CCPA) and a focus on **Employee Morale**.
- **Final Thought:** "In the digital age, the most successful organizations are those that manage their data as carefully as their capital, and their people as humanely as their purpose

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THANK YOU!